Measures for Model Performance and Evaluation

A comprehensive guide to Classification Metrics

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Objectives

- Understand key performance metrics for classification models
- Learn to interpret confusion matrices
- Evaluate trade-offs between different metrics
- Select appropriate metrics for different business scenarios

Introduction

In machine learning, an evaluation metric is a way used to measure how well a model's predictions align with the actual outcomes. It helps in determining the model's performance by quantifying the difference between predicted and true values.

- **In Regression problems:** The evaluation metrics measure ho accurately the model predicts continuous values.
- **In classification problems:** The evaluation metrics focus on how well the model assigns labels to data points.

Why Model Evaluation Matters

- All models make mistakes - but which mistake matters most
- Model Selection depends on evaluation metrics
- Ensures reliability and trust in predictive systems

Example Scenarios:

- Medical Diagnosis (Cancer Detection)
- Fraud Detection in banking

The Confusion Matrix

- A matrix used to determine the performance of classification models for a given set of test data.
- The matrix itself can be easily understood, but the related terminologies may be confusing.

	Predicted: No	Predicted: Yes
Actual: No	True Negative (TN)	False Positive (FP)
Actual: Yes	False Negative (FN)	True Positive (TP)

Key Components of Confusion Matrix

- **True Positive (TP):** The actual result is positive and the model also predicted that result is positive.
- **True Negative (TN):** The actual result is negative and the model also predicted that result is negative.
- **False Positive (FP):** The actual result is negative but the model predicted that result is positive. (Type I error)
- **False Negative (FN):** The actual result is positive but the model predicted that result is negative. (Type II error)

Accuracy

- It defines how often the model predicts the correct output.

<u>**Accuracy**</u> = TP + TN / TP + TN + FP + FN

<u>**When to Use:**</u>

- Balanced Classes (Similar no. of positive and negative classes)
- When FP and FN have similar costs

Misclassification Rate (Error Rate)

- It defines how often the model gives the wrong predictions.

Error Rate= FP + FN / TP + TN + FP + FN

When to Use:

- Accessing the overall accuracy of a model where the costs of different types of errors (False Positives vs. False Negatives) are roughly equal and the dataset is balanced.

Precision (Positive Predictive Value)

- The number of correct outputs provided by the model or out of all positive classes that have predicted correctly by the model.

Precision = TP / TP + FP

When Precision Matters:

- Spam Detection
- Search engine results
- Quality over quantity scenarios

Sensitivity/Recall (TPR, Hit Rate)

- It is defined as the out of total positive classes, how our model predicted correctly. The recall must be as high as possible.

Recall = TP / TP + FN

When Recall Matters:

- Disease screening
- Fraud Detection
- Safety-critical applications

Specificity (TNR, Selectivity)

- It is classification metric which measures the proportion of actual negative cases correctly identified as negative.

Specificity = TN / TN + FP

When Specificity Matters:

- When false positives are particularly harmful
- Legal screening systems
- High-stakes negative predictions

F1 Score (F-Measure)

- It is classification metric that helps to evaluate the recall and precision at the same time.

F1-Score = 2 * Precision * Recall / Precision + Recall

When F1-Score Matters:

- Imbalance Datasets
- When both false positives and false negative matters
- Comparing models with different precision/ recall trade-offs

Some Other Parameters

- <u>Negative Predictive Value (NPV)</u> = TN / TN + FN
- <u>False Negative Rate (FNR)</u> = 1 - Sensitivity
- <u>False Positive Rate (FPR)</u> = 1 - Specificity
- <u>False Discovery Rate (FDP)</u> = 1 - Precision
- <u>False Omission Rate (FOR)</u> = 1 - NPV

The Precision-Recall Tradeoff

The inevitable tradeoff:

- Inverse relation
- Determined by classical threshold

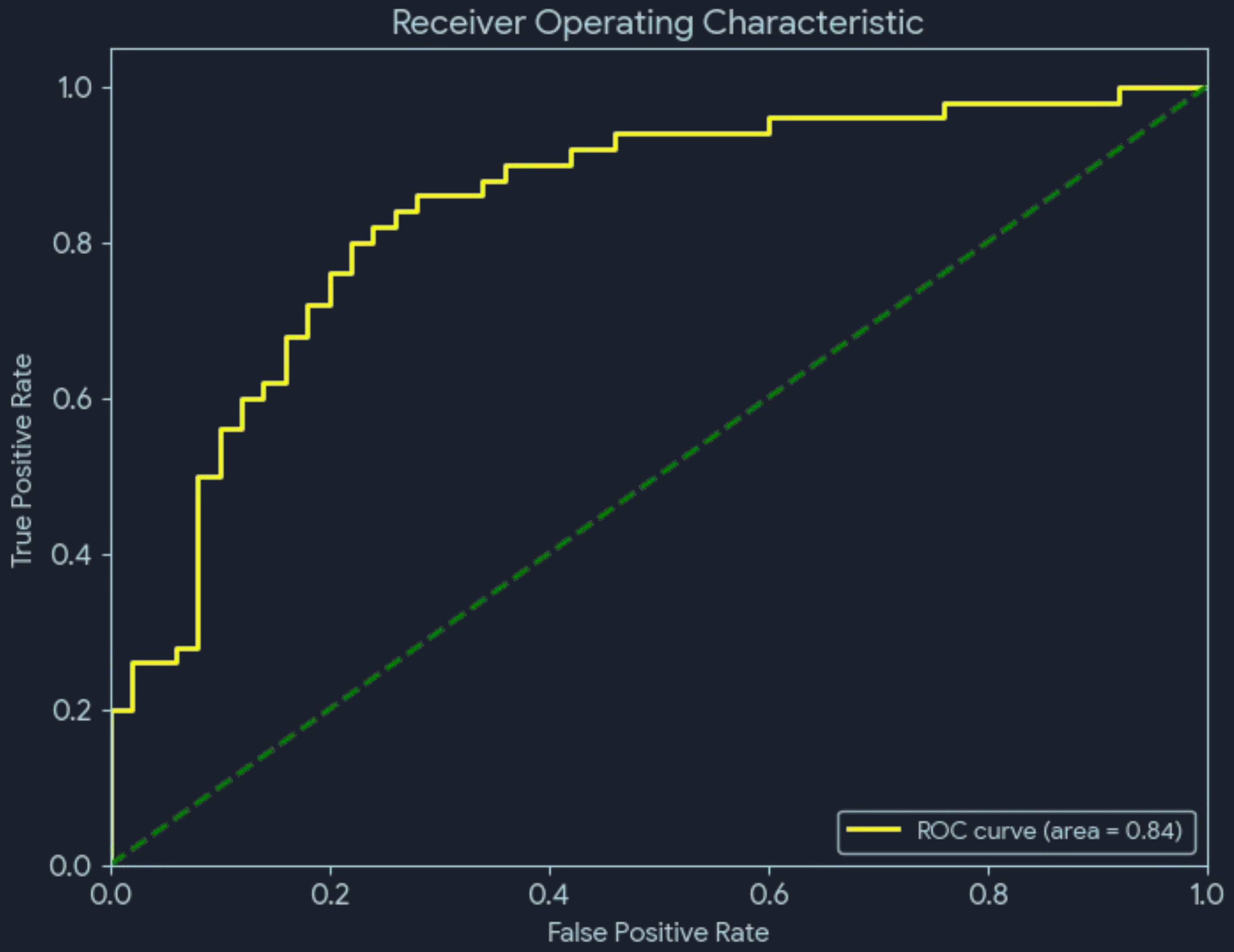
Threshold Adjustment:

- Higher Threshold -> Higher precision, lower recall
- Lower Threshold -> Higher recall, lower precision

AUC-ROC

The Receiver Operator Characteristics (ROC) is a probabilistic curve that plots the TPR (True Positive Rate) against the FPR (False Positive Rate) at various threshold values and separates the 'signal' from the 'noise'.

The Area Under Curve (AUC) is the measure of the ability of a classifier to distinguish between classes.


Receiver Operating Characteristic
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC curve (area = 0.84)

<u>The Area Under Curve (AUC)</u> refers to the area covered by the ROC.

- AUC = 1 -> Perfect classifier
- AUC > 0.5 -> Better than random guess
- AUC = 0.5 -> Random guessing
- AUC < 0.5 -> Worse than random guess

<u>The threshold</u> is the decision point that determines the classification boundary between positive and negative classes. (Generally 0.5)

Optimal Threshold Selection: While the ROC curve shows all thresholds, the "optimal" threshold is often selected by finding the point on the curve closest to the top-left corner (highest TPR, lowest FPR), sometimes using metrics like the Youden's J statistic (J = Sensitivity + Specificity – 1) or maximizing the F1-score.

Log Loss also called Cross-Entropy loss is one of the major metrics to access the performance of a classification problem. For a single sample with true label y (0,1) :-

$$\text{Log loss} = y * \log(p) + (1 - y) * \log(1 - p)$$

Where, p = P(y = 1) is probability estimate

Numerical Example

Model	Predicted Yes	Predicted No
Actual Yes	40	60
Actual No	80	20

Solution:

Given,

TP = 40

TN = 60

FP = 80

FN = 20

Accuracy = TP + TN / TP + TN + FP + FN

= (40 + 60) / (40 + 60 + 80 + 20) = 0.5 = 50%

Precision = TP / TP + FP

= 40 / (40 + 80) = 0.33 = 33.33%

Recall = TP / TP + FN

= 40 / (40 + 20) = 0.67 = 67%

F1-Score = 2 * Precision * Recall / Precision + Recall

= (2 * 0.33 * 0.67) / (0.33 + 0.67) = 0.44

= 44%

Practical Example (Medical Diagnosis)

Scenarios: Cancer screening test

- Positive = Cancer detected
- Negative = No cancer detected

Comparison of two models:

Metric	Model A	Model B
Accuracy	92%	90%
Precision	85%	95%
Recall	70%	60%
F1-Score	77%	74%

Choosing the Right Metric

Decision Framework:

- **Imbalance Data?** -> Use F1, Precision, Recall instead of Accuracy
- **False positives correctly?** -> Prioritize Precision
- **False negatives correctly?** -> Prioritize Recall
- **Need Balance?** -> Use F1-Score
- **Both types of error matter?** -> Consider cost matrix

Choosing the Right Metric

Industry Examples:

- **Finance (Fraud):** High Precision (Minimizes false accusations)
- **Healthcare:** High recall (don't miss diagnoses)
- **Marketing:** Balance based on campaign costs

Golden Rule: *No single metric tells the whole story - choose metrics based on your specific problem and business requirements.*

Summary and Key Takeaways

<u>**Core Metrics:**</u>

1. **Confusion Matrix:** Foundation for all metrics
2. **Accuracy:** Overall correctness (caution with imbalance)
3. **Precision:** Quality of positive predictions
4. **Recall/Sensitivity:** Ability to find positive cases
5. **F1 Score:** Balance between precision and recall
6. **Log Loss:** Model confidence when correct predictions

Q & A and Further Resources

Questions??

Recommended Resources:

- Scikit-Learn documentation on metrics
- "Hands-On Machine Learning" by Aurelien Geron
- Coursera: "Machine Learning" by Andrew Ng
- Towards Data Science articles on evaluation metrics

Thank You!!!